

Using common Maclaurin's series

Basic skills

$$Q1. a) \sin x = x - \frac{x^3}{3!} + \frac{x^5}{5!} - \dots$$

$$b) \cos x = 1 - \frac{x^2}{2!} + \frac{x^4}{4!} - \dots$$

$$c) e^x = 1 + x + \frac{x^2}{2!} + \frac{x^3}{3!} + \dots$$

$$d) \ln(1+x) = x - \frac{x^2}{2} + \frac{x^3}{3} - \dots$$

$$Q2. \cdot \ln(1+x) \quad -1 < x \leq 1$$

$$\cdot \arctan x \quad -1 < x \leq 1$$

$$\cdot (1+ax)^n \quad |x| \leq 1$$

sometimes
hidden
in binomial
expansion

$$Q3. e^{2x} = 1 + (2x) + \frac{(2x)^2}{2} + \frac{(2x)^3}{6} + \dots$$
$$= 1 + 2x + 2x^2 + \frac{8x^3}{6} + \dots$$

$$Q4. \sin\left(\frac{x}{2}\right) = \left(\frac{x}{2}\right) - \frac{\left(\frac{x}{2}\right)^3}{3!} + \dots$$
$$= \frac{x}{2} - \frac{x^3}{48} + \dots$$

Competency skills

Q5. $e^{\sin x} = 1 + \sin x + \frac{\sin^2 x}{2} + \frac{\sin^3 x}{6} + \dots$
x → sin x in e^x

$$= 1 + \left(x - \frac{x^3}{3!} + \dots\right) + \frac{\left(x - \frac{x^3}{3!}\right)^2}{2} + \frac{\left(x - \frac{x^3}{3!}\right)^3}{6} + \dots$$
$$= 1 + x - \frac{x^3}{6} + \frac{x^2}{2} + \frac{x^3}{6} + \dots$$
$$= 1 + x + \frac{x^2}{2} + \dots$$

x → x²

Q6. a) $\cos(x^2) = 1 - \frac{(x^2)^2}{2} + \dots = 1 - \frac{x^4}{2}$

b) $\sin(x^2) = x^2 - \frac{(x^2)^3}{6} + \dots = x^2 - \frac{x^6}{6}$

c) $\tan(x^2) = \frac{\sin(x^2)}{\cos(x^2)}$

$$= \sin(x^2) (\cos(x^2))^{-1}$$

$$= \left(x^2 - \frac{x^6}{6}\right) \left(1 - \frac{x^4}{2}\right)^{-1}$$

binomial exp

$$= \left(x^2 - \frac{x^6}{6}\right) \left(1 - 1 \times -\frac{x^4}{2} + \dots\right)$$

$$= x^2 + \frac{x^6}{2} - \frac{x^6}{6}$$

$$= x^2 - \frac{x^6}{3} + \dots$$

$x \rightarrow \sin x$

Q7. a) $f(x) = \ln(1 + \sin x)$

$$= \sin x - \frac{(\sin x)^2}{2} + \frac{(\sin x)^3}{3} - \dots$$

$$= \left(x - \frac{x^3}{6} + \dots\right) - \frac{\left(x - \frac{x^3}{6}\right)^2}{2} + \frac{\left(x - \frac{x^3}{6}\right)^3}{3}$$

$$= x - \frac{x^3}{6} - \frac{x^2}{2} + \frac{x^3}{3}$$

$$= x - \frac{x^2}{2} + \frac{x^3}{6}$$

$x \rightarrow \sin x$ so

all troughs

b) $-1 < \sin x \leq 1$
so $x \neq \frac{3\pi}{2} + 2\pi n$ for $n \in \mathbb{Z}$

$$c) f\left(\frac{\pi}{3}\right) \approx \frac{\pi}{3} - \frac{\left(\frac{\pi}{3}\right)^2}{2} + \frac{\left(\frac{\pi}{3}\right)^3}{6} = 0.69028\dots$$

$$f\left(\frac{\pi}{3}\right) = \ln\left(1 + \sin\frac{\pi}{3}\right) = 0.62381\dots$$

$$\frac{0.69028\dots - 0.62381}{0.62381}$$

$$= 0.107 = 10.7\%$$

Advanced skills

Q8. Because we can not find $f(0), f'(0), \dots$

As undefined at $x=0$

Q9. $\ln(1+e^x) = e^x - \frac{(e^x)^2}{2} + \frac{(e^x)^3}{3}$

might try $x \rightarrow e^x$

$$= 1 + x + \frac{x^2}{2} - \frac{\left(1 + x + \frac{x^2}{2}\right)^2}{2} + \frac{\left(1 + x + \frac{x^2}{2}\right)^3}{3} + \dots$$

However need infinity as

$(1+x+\dots)^n$ always has

Some units

Instead

$$\ln(1+e^x) = \ln\left(1+1+x+\frac{x^2}{2}+\frac{x^3}{6}+\dots\right)$$

$$= \ln\left(2+x+\frac{x^2}{2}+\dots\right)$$

$$= \ln 2 + \ln\left(1+\frac{x}{2}+\frac{x^2}{4}+\dots\right)$$

$$= \ln 2 + \left(\frac{x}{2} + \frac{x^2}{4}\right) - \frac{\left(\frac{x}{2} + \frac{x^2}{4}\right)^2}{2} + \dots$$

$x \rightarrow \frac{x}{2} + \frac{x^2}{2} + \dots$

$$= \ln 2 + \frac{x}{2} + \frac{x^2}{4} - \frac{\frac{x^2}{4}}{2} = \ln 2 + \frac{x}{2} + \frac{x^2}{8}$$

valid for

$$-1 < \frac{x}{2} + \frac{x^2}{4} + \dots \leq 1$$

$$-2 < x + \frac{x^2}{2} + \dots \leq 2$$

$$-1 < 1 + x + \frac{x^2}{2} + \dots \leq 1$$

$$-1 < e^x \leq 1$$

as $e^x > 0$
all -ve x
work

$$-\infty < x \leq 0$$

$$\nrightarrow \ln 1 = 0$$

Q10. $\ln(e^{-3x} + e^{3x})$

$$= \ln\left(1 - 3x + \frac{(-3x)^2}{2} + \dots + 1 + 3x + \frac{(3x)^2}{2}\right)$$

$$= \ln\left(1 + \frac{9}{2}x^2 + \frac{9}{2}x^2 + \dots\right) = \ln(2 + 9x^2 + \dots)$$

$$= \ln 2 + \ln\left(1 + \frac{9}{2}x^2 + \dots\right)$$

laws of logs

$$= \ln 2 + \frac{9}{2}x^2 + \dots$$

Q11. $\sin(x^2) = x^2 - \frac{(x^2)^3}{6} + \frac{(x^2)^5}{120}$

So

$$\frac{\cancel{x^2} - \frac{\cancel{x^6}}{6} + \frac{x^{10}}{120} - \dots - \cancel{x^2} + \frac{1}{6}\cancel{x^6}}{x^{10}(x^5 + 5)}$$

$$= \frac{\frac{\cancel{x^{10}}}{120} - \dots}{\cancel{x^{10}}(x^5 + 5)} = \frac{1 - \dots}{120(x^5 + 5)}$$

as $x \rightarrow 0 \rightarrow \frac{1}{120 \times 5} = \frac{1}{600}$

Q12.

$$e^{3x} \cos 2x = \left(1 + 3x + \frac{(3x)^2}{2} + \dots\right) \left(1 - \frac{(2x)^2}{2} + \dots\right)$$

$$= \left(1 + 3x + \frac{9x^2}{2} + \dots\right) \left(1 - 2x^2 + \dots\right)$$

$$= 1 - 2x^2 + 3x + \frac{9x^2}{2}$$

$$= 1 + 3x + \frac{5}{2}x^2$$

$$(1+ax)^n = 1 + nax + \frac{n(n-1)}{2}(ax)^2$$

$$\text{So } na = 3 \quad \frac{n(n-1)}{2}a^2 = \frac{5}{2}$$

$$a = \frac{3}{n} \Rightarrow \frac{n(n-1)}{2} \times \frac{9}{n^2} = \frac{5}{2}$$

$$\text{So } \frac{9(n-1)}{n} = 5 \quad 9n - 9 = 5n$$

$$4n = 9 \quad n = \frac{9}{4}$$

$$a = \frac{3}{\frac{9}{4}} = \frac{12}{9} = \frac{4}{3}$$